The Reality of AI-Assisted and AI-Based Cyber Attacks: Myths and Realities, a Deep Dive

Item	Detail
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Evidence reviewed to	7 November 2026 (latest sources: Anthropic and Google threat reports of 10 and 8 September 2026)
Companion file	AI_Cyber_Attacks_Myths_and_Realities.pptx (presentation of this document)
Safety note	This is a defensive and educational analysis. It contains no instructions for building or running attacks, and the indicators in vendor reports are not reproduced.
Citations	Numbers in square brackets refer to the reference list in section 12.

1. Executive summary

AI is now a routine part of cyber attacks, but the popular picture of it is wrong in both directions. It is not science fiction: since late 2025, threat reports from Anthropic, Google and OpenAI document real intrusions, extortion, espionage and fraud in which AI wrote tools, ran reconnaissance, executed commands and processed stolen data [1][2][4][6]. It is also not magic: the techniques are mostly familiar, humans still choose targets and decide what to monetize, AI makes mistakes, and the best-known statistics are sometimes wrong or unverifiable [1][7][13][14].

The most defensible one-sentence conclusion is this: **AI has changed the economics of attacks (speed, scale and the skill needed), not the basic kinds of attack.** Attackers still win mainly through stolen credentials, unpatched internet-facing systems, phishing and impersonation [1][8]. What changed is that a lone operator can now do work that once needed a team, that the gap between a patch and an exploit keeps shrinking, and that AI systems, API keys and developer tooling have become targets in their own right [1][4][7].

Headline finding	Confidence	Main evidence
AI is used across the whole attack lifecycle by state and criminal actors	High	Google, Anthropic, OpenAI reports; UK NCSC assessment [1][4][5][6][7]
Agentic (multi-step, tool-using) AI now runs parts of real intrusions with limited human supervision	Medium to high	Anthropic cases from 2025 and 2026; Google observed an agent-enabled campaign built in under six hours [1][2][4]
Fully autonomous, end-to-end attacks that choose their own targets are not yet confirmed in the wild	Medium	Google reports none observed; Anthropic notes humans keep target selection and review [1][4]
AI-written or AI-orchestrated malware is mostly experimental so far	Medium	PromptLock was an academic prototype; LameHug and HONESTCUE are early experiments [5][15][16][17]
Claims such as '80% of ransomware is AI-driven' are unsupported	High	The paper was withdrawn after expert criticism [13]
Basic security hygiene still matters most	High	Verizon DBIR 2026; Anthropic and NCSC conclusions [1][7][8][9]
Defenders gain similar AI capability, with a head start in some programmes	Medium	Project Glasswing results; Google AI defence tools [4][11]

2. Scope, definitions and how to read the evidence

2.1 Terms used in this document

Term	Meaning here
AI-assisted attack	A human attacker uses AI as a helper: drafting lures, summarizing data, writing or debugging code, researching targets.
AI-based or AI-enabled tooling	The attack toolkit itself contains AI, for example malware that calls a language model at run time to generate commands or code.
Agentic or AI-orchestrated attack	AI agents plan and carry out multi-step tasks using tools (scanners, shells, browsers) with the human giving goals and reviewing results.
Vibe hacking	Anthropic's label for giving an AI a general goal and letting it assess the environment, write and run scripts and repeat until done [3].
Attacks on AI	Targeting AI systems and their ecosystem: stolen API keys, prompt injection, poisoned tools, model theft (distillation), hijacked compute [1][4].

2.2 A practical autonomy scale

The reports describe a wide range of autonomy. The scale below is my own summary of those descriptions, offered to make debates precise; it is not an official standard.

Level	What the AI does	Evidence status
L0	Chat assistant: answers questions, drafts text and code	Widespread [5][6][7]
L1	Writes and refines tools or malware under human direction	Widespread [4][5]
L2	Runs commands against victims while a human decides each target and step	Documented [1]
L3	Multi-agent frameworks run reconnaissance, exploitation and theft across several victims; humans set targets and review	Documented by Anthropic and Google [1][2][4]
L4	End-to-end autonomy including choosing targets and objectives	Not confirmed in the wild; Google says not yet observed [4]

2.3 How reliable is each kind of source?

Source type	Strength	Limitation
Platform threat reports (Anthropic, Google, OpenAI)	Direct telemetry on real misuse; detailed case studies	Self-reported; they see only their own platforms; case selection is not a random sample [1][4][6]
Government assessments (UK NCSC)	Independent, uses probability language, forward-looking	Broad; assessments age quickly [7]
Incident datasets (Verizon DBIR, M-Trends)	Large samples of real breaches	AI involvement is coded coarsely [8][9][10]
Academic prototypes	Show what is technically possible	Feasibility is not prevalence [15][16]

Source type	Strength	Limitation
Vendor blogs and statistics roundups	Fast, sometimes useful pointers	Methods often unclear; treat as unverified until traced to a primary source [23]

3. Evidence timeline, 2024 to 2026

Date	Event or source	Why it matters
Jan 2024	Arup deepfake video-call fraud, about USD 25.6 million over 15 transfers [22][23]	Proof that deepfake impersonation can defeat normal checks
May 2025	UK NCSC: AI will almost certainly raise the volume and impact of intrusions to 2027 by enhancing existing methods [7]	Official baseline: evolution, not new attack types
Jul to Aug 2025	CERT-UA reports LameHug (attributed to APT28); ESET reports PromptLock, later shown to be an NYU research prototype [15][16][17]	First LLM-driven malware, mostly experimental
Aug 2025	Anthropic describes vibe-hacking extortion against at least 17 organizations [3]	AI as operator, not just advisor
Nov 2025	Anthropic reports GTG-1002, a state-linked campaign against about 30 targets using agentic AI [2]	First widely cited AI-orchestrated espionage case; also heavy debate [14]
Nov 2025	MIT Sloan withdraws the '80% of ransomware uses AI' paper [13]	Warning about hype and weak statistics
Feb 2026	Google GTIG tracker and OpenAI report: AI used for recon, phishing, coding, with early AI-integrated malware [5][6]	Cross-vendor confirmation
Apr to May 2026	Project Glasswing: frontier model finds over 10,000 high or critical flaws in the first month [11]	Defensive use of the same capability
May 2026	Verizon DBIR 2026: exploitation overtakes stolen credentials as top initial access vector [8][9]	Speed of exploitation is now the central problem
Sep 2026	Google GTIG 'From Prompting to Autonomy' and Anthropic's September 2026 report [1][4]	Latest evidence of agentic operations and attacks on AI supply chains

4. How attackers use AI across the attack lifecycle

The table summarizes what the latest reports show at each stage, with a maturity judgment of my own: Mature means routine and widespread, Growing means increasingly common, Emerging means documented in a few cases, Contested means evidence conflicts.

Stage	Observed AI use	Maturity	Sources
Reconnaissance	Profiling people and organizations, fingerprinting exposed services, building target lists	Mature	[1][4][5]
Social engineering	Localised lures, fake personas and resumes, translation, phishing infrastructure	Mature	[1][4][5]
Initial access	Prototyping n-day exploits after patches, mass credential harvesting, device-code phishing platforms	Growing	[1][4]
Malware and tooling	Writing, obfuscating and debugging tools; phishing kits; run-time LLM calls in experimental malware	Growing, experimental for run-time LLM malware	[4][5][17]
Post-exploitation	Running commands, lateral movement, token and data theft, sorting stolen data	Emerging to growing	[1][4]
Defence evasion	Agents that rebuild tools when security products flag them	Emerging	[1]
Zero-day discovery	Frontier models find many flaws in controlled defensive programmes; criminal use is reported but disputed in scope	Contested	[1][4][11]
Extortion and fraud	Processing stolen data, deepfake voice and video, fake storefronts	Mature	[1][22][23]
Influence operations	Content factories and personas; most operations reached few real people	Mature but low impact	[1][4]
Attacks on AI itself	API key theft, prompt injection, poisoned tools, distillation, hijacked compute	Growing	[1][4][19]

5. Deep dives into the key cases

5.1 GTG-1002: an AI-orchestrated espionage campaign (September 2025)

Anthropic reported in November 2025 that a group it assessed with high confidence as Chinese state-sponsored used its coding agent, connected to common open-source security tools through the Model Context Protocol, against roughly 30 organizations in technology, finance, chemicals and government. Anthropic estimated that the AI performed 80 to 90 percent of the tactical work, with humans involved for about 10 to 20 percent, mostly approving escalations and reviewing data before theft. Some intrusions succeeded [2][14].

Two points make the case instructive. First, the operators did not need new malware; they built a framework that broke the work into small tasks and persuaded the model it was assisting legitimate security testing. Second, the model made mistakes: it sometimes overstated findings or claimed credentials that did not work, so the operators had to validate its output [2]. Critics noted that no technical indicators were published at the time, that only some attacks succeeded, and that much of the automation resembled existing tools [14]. Both the significance and the caveats are real. Later reporting says this operating model has since spread to many other actors [1].

5.2 Vibe hacking and agent swarms (2025 to 2026)

Anthropic's August 2025 report described an extortion operation that used an AI coding agent for reconnaissance, credential harvesting and network entry against at least 17 organizations, including healthcare and emergency services [3]. Its September 2026 report, covering December 2025 to August 2026, extends the picture with several cases [1][18]:

Actor label	What was reported	Lesson
GTG-50014 (suspected ShinyHunters affiliates)	Credential-harvesting pipelines (for example scanning 1.8 million Android apps for hardcoded secrets); breaches taking hours; one case from a single stolen token to cloud admin in about three hours; thousands of downstream tenants reached via a SaaS supplier	AI scales opportunistic crime; obscurity is no longer protection
GTG-10007 (Chinese-speaking operators)	Agent swarms with persistent memory against about 50 organizations; an automated vulnerability-research programe against security appliances that produced more than a dozen possible zero-day findings in one month	Skilled labor and exploit supply are being automated
GTG-20006 (consistent with Russian state espionage)	AI-driven workflows for phishing, infrastructure and exfiltration against more than 20 organizations; agents rebuilt malware when it was detected	Static detection lose value; speed of response matters
GTG-50029 (single French-speaking hacktivist)	Used stolen API keys and AI agents against 42 targets and reached at least 14; built a mass doxxing platform alone	Small actors can now run state-level campaigns
GTG-50020	Attacked about 30 AI companies in roughly four days to reach a pre-release model; every attempt failed	AI vendors and their customers are targets

Anthropic draws two cautions that should travel with any summary. Humans still made the decisions that matter most to them (targets, monetization, review), and autonomy and harm are separate: several of the most damaging compromises came from operations where a human directed every step. Also, none of the 2026 misuse cases used its most capable restricted models except one distillation case, and the report states the cases were the most notable it found, not typical misuse [1].

5.3 Google's view: from prompting to automation (2026)

Google's Threat Intelligence Group describes a steady shift from simple prompting to agentic workflows. In Q2 2026, Mandiant observed an actor compromise a cloud resource and then plan, build and run an agent-enabled mass credential-harvesting campaign in under six hours, compromising thousands of third-party credentials. A separate exposed server held an automated framework managing more than 23,800 harvested secrets [4]. At the same time Google reports it has not yet seen fully autonomous exploitation pipelines used against targets in the wild, and describes adversary use as maturing gradually rather than leaping ahead [4]. Its February 2026 report likewise found no breakthrough capability among tracked state actors, only efficiency gains [5].

These statements are compatible with Anthropic's if read carefully: agentic operations with human target-setting are documented, while end-to-end autonomy including target choice is not. Because each vendor sees only its own platform, neither view is complete.

5.4 Exploits and zero-days: where the real race is

Google reports actors using commercial and open-weight models to turn public patch information into working n-day exploit code, including a case about a month after a Firefox patch, and notes that shared 'knowledge files' on underground forums were largely impractical rather than ready exploits [4]. The UK NCSC judged in 2025 that the time between disclosure and exploitation had already shrunk to days and that AI would almost certainly reduce it further, while zero-day discovery gains would mainly go to skilled, well-resourced actors [7].

On the defensive side, Anthropic's restricted Project Glasswing gave about 50 organizations access to an unreleased frontier model and reported more than 10,000 high or critical vulnerabilities found in the first month; Mozilla fixed 271 vulnerabilities in Firefox 150 after testing it, against 22 security-sensitive bugs found earlier with a previous model [11]. The dual-use point is explicit: Anthropic restricted release because the same ability could help attackers build exploits [11][12]. The practical message is that patching speed, not discovery, becomes the bottleneck. Verizon's 2026 data already shows exploitation as the leading initial access vector at 31 percent, with only 26 percent of critical known-exploited vulnerabilities fully remediated and a median remediation time of 43 days [8][9].

5.5 AI systems, keys and developer tools as targets

Google tracked a cybercrime cluster (UNC6780) conducting open-source supply chain compromises across package registries and building features into its malware that interact with AI coding tools: hiding files in AI-assistant project folders, using configuration files to instruct assistants to run commands, and embedding adversarial text in scripts to make LLM security scanners refuse to analyse them [4]. Anthropic describes criminal markets for stolen AI keys and accounts, fake 'discount' AI resellers that harvest credentials, prompt injection against AI vendors' evaluation sandboxes to steal production keys, and actors who switch their own workloads onto victims' keys [1]. Google reports that underground prices for AI accounts more than doubled in 2026, and distillation campaigns against its models exceeding 100 million prompts [4].

The OWASP 2026 lists reflect this shift: prompt injection stays first, and excessive agency (giving an AI too many tools, permissions or autonomy) rose to third place [19][20][21].

5.6 Deepfakes, voice cloning and AI-assisted fraud

The Arup case in January 2024 remains the benchmark: a finance employee in Hong Kong joined a video call in which every other participant was a deepfake and made 15 transfers worth about USD 25.6 million [22][23]. Later statistics should be handled with care. Secondary summaries cite voice phishing as the second most common initial access route in 2025 and the most common into cloud environments (23 percent) per Mandiant's M-Trends 2026, and about USD 893 million in AI-enabled fraud losses recorded by the FBI for 2025, its first year of tracking the category; these came from a vendor roundup rather than the primary reports, so confirm them against M-Trends and the FBI IC3 report before quoting [23]. Claims that humans can almost never spot deepfakes, or that AI phishing gets 'four times' the clicks, circulate widely but I could not trace them to a verified primary source in this review, so they are treated as unverified.

The strongest defense is process, not detection skill: out-of-band confirmation for payments and credential resets, because a convincing face or voice is no longer proof of identity.

5.7 AI-enabled malware: experiments more than epidemics

Example	What it is	Reality check
LameHug (Jul 2025)	CERT-UA reported malware, attributed to APT28, that asks a hosted LLM to generate Windows commands [17]	Early state experiment; code was simple

Example	What it is	Reality check
PromptLock (Aug 2025)	ESET called it the first AI-powered ransomware; it used an open-weight model to generate scripts [15]	Later shown to be NYU's 'Ransomware 3.0' research prototype, non-functional outside the lab [15][16]
HONESTCUE (2026)	Google reports a downloader that uses a model's API to receive code for a second stage [5]	Experimentation; no sign of broad deployment
Agentic toolkits (2026)	Public offensive-agent frameworks such as PentAGI reproduce much of the scaffolding [1]	Lowers the barrier; mainly automates known techniques

Run-time LLM calls make signatures less useful and can hide logic from static analysis, but they also add network traces, API keys and reliability problems for the attacker. Anthropic's finding that an actor automatically rebuilt detected tools is more worrying than any single AI-powered sample, because it shortens the defender's cycle of detect, block, adapt [1].

6. Myths and realities

Verdicts: Myth means the claim is not supported; Mostly myth means it is mostly wrong with a small true element; Partly true means a real kernel with important limits; Unverified means I could not trace the claim to a reliable source.

Common claim	Verdict	What the evidence shows	Refs
AI hackers are fully autonomous and need no humans	Mostly myth	Agentic operations are real, but humans set targets, review results and monetize; models hallucinate; Google has not seen fully autonomous pipelines in the wild	[1][2][4]
AI has created entirely new kinds of attack	Mostly myth	NCSC and Anthropic both find familiar techniques (stolen credentials, unpatched devices, phishing) made cheaper and faster	[1][7]
80 percent of ransomware attacks use AI	Myth	The source paper was withdrawn after experts said it labelled ordinary ransomware groups as AI-driven without evidence	[13]
Only nation-states can run sophisticated AI campaigns	Myth	Anthropic reports lone operators, criminals and hacktivists running multi-victim campaigns once needing teams; sophistication no longer identifies the actor	[1]
AI malware that rewrites itself makes defence hopeless	Partly true	Most run-time LLM malware is experimental; the real change is automated rebuild-and-redeploy loops that make static signatures weaker	[1][5][15][16]
AI will flood the world with zero-days overnight	Partly true	Frontier models do find vulnerabilities at scale, mainly in controlled defensive programs; in-the-wild gains so far are faster n-day exploitation	[4][7][11]

Common claim	Verdict	What the evidence shows	Refs
Model safeguards stop misuse	Partly true	Providers disrupt many operations, but actors use jailbreak-style framing, stolen keys, fake resellers and open-weight models, so safeguards reduce rather than remove risk	[1][4]
AI-written phishing is undetectable and gets four times the clicks	Unverified	AI improves personalization and scale; headline click-rate figures lack traceable methods in this review	[4][23]
Deepfakes make verification impossible	Mostly myth	Real losses occur, but call-back and dual-approval procedures defeat them; the problem is skipping the procedure	[22][23]
Attacks on AI systems are only theoretical	Myth	Key theft, prompt injection of scanners and vendors, trojanised tools and distillation are documented in 2026	[1][4][19]
AI makes traditional security obsolete	Myth	Exploited vulnerabilities, stolen credentials and the human element (62 percent of breaches) still dominate the data	[8][9]
Defenders cannot use AI the same way	Myth	Glasswing, Google's defense tools and national tools such as NCCIA's Katalyst show defensive uptake	[4][11][28]
AI threat reports are just marketing	Partly true	Early scepticism was justified (no indicators at first), but later reports add indicators and multiple vendors and governments report similar patterns	[1][5][6][7][14]

7. What the numbers say: the Verizon 2026 DBIR

Verizon's 2026 report analysed more than 22,000 confirmed breaches (period November 2024 to October 2025). Reported findings relevant here [8][9][10]:

Finding	Why it matters for AI-enabled attacks
Vulnerability exploitation caused 31 percent of breaches, above credential abuse at 13 percent	Speed of exploitation, which AI helps, is now the leading route in
Only 26 percent of critical known-exploited vulnerabilities were fully remediated; median time rose from 32 to 43 days	The gap between exploit and patch is where AI helps attackers most
The median threat actor used generative AI across 15 ATT&CK techniques, some across 40 to 50	AI is a standard part of tradecraft, mostly accelerating known techniques
The human element featured in 62 percent of breaches; 48 percent involved a third party	People and suppliers remain the main weaknesses
45 percent of employees regularly use AI on corporate devices; 67 percent of AI use was via non-corporate accounts	Shadow AI creates data-leak risk on the defender's side

Caveat: DBIR codes AI involvement from partner data, so these figures show direction and rough scale, not precise measurement.

8. What defenders should do

The consistent advice across NCSC, Verizon and the vendor reports is to strengthen fundamentals and to assume faster attackers: hours, not weeks [1][7][8]. The table maps each control to the threat it addresses.

Control	Threat addressed	Evidence
Patch exploited vulnerabilities first; shrink internet-facing exposure, especially edge devices	Fast n-day exploitation, opportunistic mass scanning	[1][4][8][9]
Phishing-resistant MFA; restrict device-code sign-in flows; conditional access	Credential and token theft, device-code phishing	[1]
Out-of-band verification for payments, credential resets and help-desk requests	Deepfake and voice-clone fraud	[22][23]
Secrets hygiene: scan code, containers and mobile apps; vault; short-lived tokens; rotation	Industrial-scale key harvesting	[1][4]
Treat AI API keys as production credentials: spend limits, anomaly alerts, buy only through official channels	Stolen keys, fake resellers, hijacked compute	[1][4]
Supply chain controls: lock and review dependencies, vet MCP servers, trusted publishing, watch AI-tool config folders	Trojanized packages and tools, prompt injection via project files	[4]
Least agency for AI agents: minimal tools and permissions, human approval for risky actions, sandboxing, full logging	Goal hijack, excessive agency	[19][20][21]
Behaviour-based detection and rapid containment; assume signatures decay	Automated evasion loops	[1]
Use defensive AI for vulnerability discovery, triage and patch drafting, with human review	Patch backlog and discovery speed	[4][11]
Govern shadow AI and data sharing with external tools	Data leakage, source code exposure	[8][9]
Forensic readiness: preserve AI API and agent logs; validate AI-claimed findings; record tool-use traces	Investigation of AI-driven intrusions	[2]

9. Context for Pakistan and for teaching

9.1 Local signals

- The NCCIA reportedly received 150,542 complaints in 2025, mostly financial fraud, and the government said it was finalizing its first national AI policy, noting that deepfake and voice-cloning scams are a fast-evolving risk [24][29].
- In January 2026 the National Assembly heard that fraudsters had cloned the Speaker's voice to demand money, alongside concern about fake traffic-fine messages and bank impersonation [24][25].
- Pakistan's National CERT issued a nationwide alert on WhatsApp account hijackings in January 2026, and the PTA issued a public advisory on deepfakes in July 2026 [26][27].
- Coverage of Google's February 2026 report noted state-linked actors researching defense-sector targets in Ukraine and Pakistan [30].
- Pakistan's own defensive AI example is NCCIA's Katalyst tool for investigating online child exploitation, launched in January 2026 [28].

These are indications from news reporting, not a measured national picture of AI-enabled attacks; no source reviewed here quantifies AI's share of Pakistani incidents.

9.2 Implications for digital forensics and DevOps teaching

- Teach students to validate AI-generated findings: models can fabricate credentials and results, and an investigator must check them [2].
- Add logging and evidence preservation for AI agents and API calls to incident-response exercises.
- Include secrets scanning, dependency pinning and least-privilege tool access for coding agents in DevSecOps labs [4][19].
- Use verification-procedure drills (call-back, dual approval) rather than deepfake-spotting contests.
- Discuss dual-use ethics: the same models find and exploit flaws [11].

10. Uncertainties and what to watch

Open question	Indicator to watch
Will fully autonomous pipelines appear in the wild?	Vendor reports stating that target selection or monetisation is no longer human-driven [1][4]
How fast does time-to-exploit fall?	Gap between patch release and public exploit use; DBIR remediation medians [7][8]
Does defensive AI keep pace?	Volume of AI-found flaws that are actually patched, not just reported [11]
How much AI hype distorts decisions?	Statistics withdrawn or challenged; reports that publish indicators and methods [13][14]
Will AI supply chain attacks grow?	Incidents involving stolen keys, trojanized MCP servers and agent prompt injection [1][4][19]

Limits of this review: most primary evidence comes from three AI companies and one threat-intelligence group, so it reflects what they can see; DBIR and some statistics are summarized from secondary articles; and the field is changing monthly.

11. Short activities for a class or workshop

Activity	Time	Instructions
Myth sort	10 min	Give groups the 13 claims in section 6, cards without verdicts; they sort into Myth, Partly true, Largely true, then compare with the table
Source grading	10 min	Pick one statistic from a news article and trace it to a primary source; note if it cannot be traced
Tabletop	20 min	A finance clerk gets a video call from the 'CFO' asking for an urgent transfer; write the verification steps that would stop it
Agent least-privilege	15 min	List the minimum tools and approvals a coding agent needs for one task and what must always need human sign-off

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Accessed 7 November 2026. Primary sources are listed first; items marked secondary summarise another report.

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